

Selected Topics Of Living Systems Seminar Further Research Proposal

Daniel Haim Breger, 316136944

Technion

(Date: January 24, 2026)

The paper leaves several clear directions to expand the work, some mentioned directly by the authors.

1 Larger Systems

The system the authors used was purposefully small to study the dynamics of a single nucleation event, but another interesting thing to explore would be how systems where multiple nucleations can happen behave. Will a very large metastable system inevitably completely collapse, or could the comet structure create geometries in the system which will stabilize the system at some point?

2 More Complex Behavior

Each particle in the system move along a single dimension with a constant velocity (either positive or negative), plus a statistical diffusion. Will more complex behavior of the particles change the results?

3 Continuous Symmetries

The paper looks specifically at the case of particles with $s = \pm 1$ spin. If the particles' polar order parameter becomes continuous, such as a polarization vector with some interaction between different values, would the behavior change?